



XLTEK Brain Monitor

User and Service Manual



Publisher's Notice



105124 Rev I
XLTEK Brain Monitor



Natus Medical Incorporated
DBA Excel-Tech Ltd. (XLTEK)
2568 Bristol Circle
Oakville, Ontario, L6H 5S1 Canada
Tel: 905-829-5300 or Fax: 905-829-5304
Toll Free (US & Canada): 800-303-0306
Technical Support Email: OTS@natus.com
Website: www.natus.com



ISO 13485 Certified

Copyright © 2016 by Excel-Tech Ltd. (XLTEK)

The information in this publication is provided for reference only. All information contained in this publication is believed to be correct and complete. Excel-Tech Ltd. shall not be liable for errors contained herein nor for incidental or consequential damages in connection with the furnishing, performance, or use of this material. All product specifications, as well as the information contained in this publication, are subject to change without notice.

This publication may contain or reference information and products protected by copyrights or patents and does not convey any license under the patent rights of Excel-Tech Ltd., nor the rights of others. Excel-Tech Ltd. does not assume any liability arising out of any infringements of patents or other rights of third parties.

All rights reserved. This document contains confidential or proprietary information of Excel-Tech Ltd. No part of this document may be reproduced or transmitted in any form or by any means without the written permission of Excel-Tech Ltd.

Excel-Tech Ltd. makes no warranty of any kind with regard to this material, including but not limited to the implied warranties of merchantability and fitness of a particular purpose.

Table of Contents

INTRODUCTION	4
PRODUCT INTENDED USE	4
USING THE MANUAL	5
MANUAL CONVENTIONS	5
BRAIN MONITOR SAFETY AND STANDARDS CONFORMITY	6
STANDARDS OF COMPLIANCE AND NORMATIVE REFERENCES	6
DECLARATION OF COMPLIANCE FOR IEC 60601-1-2	8
DECLARATION OF COMPLIANCE FOR FCC	12
WARNINGS AND CAUTIONS	13
GENERAL WARNINGS	13
ELECTRICAL WARNINGS AND CAUTIONS	14
PATIENT ENVIRONMENT WARNINGS AND CAUTIONS	14
MASIMO PULSE OXIMETER WARNINGS	15
MASIMO PULSE OXIMETER SENSOR WARNINGS	16
DESCRIPTION OF SYMBOLS	17
SPECIFICATIONS: BRAIN MONITOR AMPLIFIER	18
PRODUCT IMAGES	21
BRAIN MONITOR BASE UNIT	21
BRAIN MONITOR BREAKOUT BOX	24
UNPACKING	25
SETTING UP	26
USB-STYLE CONNECTION	26
ETHERNET-STYLE CONNECTION	27
TESTING THE BRAIN MONITOR AMPLIFIER	27
CALIBRATION AND VERIFICATION	27
CHANNEL TEST	28
IMPEDANCE CHECK	29
PULSE OXIMETER	29
INDICATIONS FOR USE	30
HOW THE PULSE OXIMETER WORKS	30
PULSE OXIMETER ACCESSORIES	32
AVAILABLE OXIMETRY SENSORS	32
MEASUREMENTS	33
MAINTENANCE AND CLEANING	34
DO'S AND DON'TS	34

<u>TROUBLESHOOTING</u>	<u>35</u>
OPENING THE TROUBLESHOOTING SECTION OF THE ONLINE HELP	36
<u>GETTING HELP</u>	<u>37</u>
<u>EEG AND SLEEP ACCESSORIES</u>	<u>38</u>

Introduction

The XLTEK Brain Monitor EEG/Sleep amplifier system is ideal for running all types of polysomnography studies along with full EEG recordings. The Brain Monitor has 50 input channels – including an integrated Masimo SET[®] pulse oximeter – and both TCP/IP and USB connectivity for quick and easy installation. This rugged device was designed with extensive clinical input to meet the workflow and application needs of the EEG/Sleep Lab.

Brain Monitor features include:

- 32 referential, 6 differential, 10 DC, and 2 oximeter channels
- Ability to initiate an impedance test, change the threshold, and view results in the patient room
- Built-in night light
- Ruggedness and durability
- TCP/IP and USB connectivity
- DC inputs on base unit and breakout box for fixed and portable devices
- Patient-event switch interface
- Photic-stimulator interface for EEG applications



WARNING: We strongly recommend that you read the **Warnings and Cautions** section of this manual before operating this amplifier.

Product Intended Use

The Brain Monitor amplifier is intended for electroencephalographic (EEG) or polysomnographic (PSG) recording. The Brain Monitor amplifier is designed to be used with **Natus** SleepWorks[™] or NeuroWorks[™] software to assist with diagnosis. It should **NOT** be used as an apnea monitor.

Using the Manual

This manual provides basic information and instructions that will enable you set up and operate the **Brain Monitor** amplifier. When reviewing the procedures, we recommend that you read the whole section first, before starting the sequence. Please follow the instructions carefully.

Manual Conventions

	This symbol denotes a warning or important information that should not be missed. Read all warnings and cautions carefully before starting the system for the first time.
	A note that contains important supplemental information.
Bold	Names of control keys, function keys, options, and labels are shown in bold. Bold text is also used to emphasize important names or ideas.
<i>Italic</i>	Italic text is used for captions.

Brain Monitor Safety and Standards Conformity

Standards of Compliance and Normative References

Sleep/EEG System, models “**Brain Monitor**” and “**ICU Brain Monitor**”; rated 115V/230V, 50/60Hz, 20W; detachable cord connected.

1. Type of protection against electric shock: Class I
2. Degree of protection against electric shock: Type BF
3. Degree of protection against ingress of water: IPX0
4. Mode of operation: Continuous
5. Environmental Conditions: Normal: 10-40°C, 30-75% rH, 700-1060hPa

The **Brain Monitor** and its accessories have been designed to comply with the following national and international standards.

Table 1 – Safety Standard of Compliance and Normative References

CAN/CSA C22.2 No 601.1-M90	Medical Electrical Equipment Part 1: General Requirements for Safety
CAN/CSA C22.2 601.1S1:1994	Supplement No 1-94 to CAN/CSA C22.2 601.1-M90 Medical Electrical Equipment Part 1: General Requirements for Safety
CAN/CSA C22.2 601.1B-90:1998	Amendment 2 to CAN/CSA C22.2 601.1-M90 Medical Electrical Equipment Part 1: General Requirements for Safety
CAN/CSA-C22.2 No. 60601-2-26-04	Medical Electrical Equipment Part 2-26: Particular Requirements for the Safety of Electroencephalographs.
UL 60601-1 (1st edition)	Medical Electrical Equipment Part 1: General Requirements for Safety
IEC 60601-1:1998	Medical Electrical Equipment Part 1: General Requirements for Basic Safety and Essential Performance
IEC 60601-2-26: 2002 (2nd Ed)	Medical Electrical Equipment Part 2-26: Particular Requirements for the Safety of Electroencephalographs
IEC 60601-1-4:2000 (1.1 Ed.)	Medical Electrical Equipment - Part 1-4: General Requirements for Safety - Collateral Standard: Programmable Electrical Medical Systems

Table 2 – EMC Standard of Compliance and Normative References

EN 60601-1-2:2001 (2nd Ed)	Medical Electrical Equipment Part 1-2:General Requirements for Safety - Collateral Standard: Electromagnetic Compatibility - Requirements and Tests
IEC 61000-4-2:1995 / EN 61000-4-2:2001	Electromagnetic Compatibility (EMC) Part 4-2: Testing and Measurement Techniques - Electrostatic Discharge Immunity Test
IEC 61000-4-3:2002 / EN 61000-4-3:2002 +A1:2002 +IS1:2004	Electromagnetic Compatibility (EMC) Part 4-3: Testing and Measurement Techniques - Radiated, Radio-frequency, Electromagnetic Field Immunity Test
IEC 61000-4-4:2004 / EN 61000-4-4:2004	Electromagnetic Compatibility (EMC) Part 4-4: Testing and Measurement Techniques - Electrical Fast Transient/Burst Immunity Test
IEC 61000-4-5:1995 / EN 61000-4-5:2001	Electromagnetic Compatibility (EMC) Part 4-5: Testing and Measurement Techniques - Surge Immunity Test
IEC 61000-4-6:1996 / EN 61000-4-6:1996 +IS1:2004	Electromagnetic Compatibility (EMC) Part 4-6: Testing and Measurement Techniques - Immunity to Conducted Disturbances, Induced by Radio-frequency Fields
IEC 61000-4-8 / EN 61000-4-8	Electromagnetic Compatibility (EMC) Part 4-8: Testing and Measurement Techniques - Power Frequency Magnetic Field Immunity Test
IEC 61000-4-11:2004 / EN 61000-4-11:2004	Electromagnetic Compatibility (EMC) Part 4-11: Testing and Measurement Techniques - Voltage Dips, Short Interruptions and Voltage Variations Immunity Tests
IEC 61000-3-2:1995 / EN 61000-3-2:2001	Electromagnetic Compatibility (EMC) Part 3-2: Limits - Limits for Harmonic Current Emissions
IEC 61000-3-3:1994 / EN 61000-3-3:1998	Electromagnetic Compatibility (EMC) Part 3-3: Limits - Limitation of Voltage Changes, Voltage Fluctuations and Flicker in Public Low-voltage Supply Systems
CISPR 11:2004 / EN 55011:1998 +A1:1999 +A2:2002 Class A/B, Group 1	Industrial, Scientific and Medical (ISM) Radio-Frequency Equipment - Electromagnetic Disturbance Characteristics - Limits and Methods of Measurement
ANSI C63.4:2004	American National Standard for Methods of Measurement of Radio-noise Emissions from Low-Voltage Electrical and Electronic Equipment in the Range of 9KHz to 40 GHz
CISPR 16-1-1:2003	Specification for Radio Disturbance and Immunity Measuring Apparatus and Methods. Part 1-1: Measuring Apparatus
CISPR 16-2-1:2004	Specification for Radio Disturbance and Immunity Measuring Apparatus and Methods. Part 2-1: Conducted Disturbances Measurement
FCC 47 CRF Part 15:2005, Subpart B Class B	Federal Communications Commission (FCC) - Unintentional Radiators
CISPR 22:2003 +A1:2004 / EN 55022:2003	Information Technology Equipment - Radio Disturbance Characteristics - Limits and Methods of Measurement

Declaration of Compliance for IEC 60601-1-2

Table 1 - Electromagnetic Emissions

Guidance and Manufacturer's Declaration – Electromagnetic Emissions		
The BRAIN MONITOR is intended for use in the electromagnetic environment specified below. The customer or the BRAIN MONITOR should assure that it is used in such an environment.		
Emissions test	Compliance	Electromagnetic Environment - Guidance
RF emissions CISPR 11	Group 1	The BRAIN MONITOR uses RF energy only for its internal function. Therefore, its RF emissions are very low and not likely to cause any interference in nearby electronic equipment
RF emissions CISPR 11	Class A	
Harmonic emissions IEC 61000-3-2	Complies	
Voltage fluctuations/ flicker emissions IEC 61000-3-3	Complies	

Table 2 - Electromagnetic Immunity

Guidance and Manufacturer's Declaration – Electromagnetic Immunity			
The BRAIN MONITOR is intended for use in the electromagnetic environment specified below. The customer or the user of the BRAIN MONITOR should assure that it is used in such an environment.			
Immunity Test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
Electrostatic Discharge (ESD) IEC 61000-4-2	±6 kV contact ±8 kV air	Complies	Floors should be wood, concrete or ceramic tile. If floors are covered with synthetic material, the relative humidity should be at least 30%.
Electrostatic fast transient/burst IEC 61000-4-4	±2 kV for power supply lines ±1 kV for input/output lines	Complies Complies	Mains power quality should be that of a typical commercial or hospital environment.
Surge IEC 61000-4-5	±1 kV differential mode ±2 kV common mode	Complies	Mains power quality should be that of a typical commercial or hospital environment.
Voltage dips, short interruptions and voltage variations on power supply input lines IEC 61000-4-11	<5% <i>UT</i> (>95% dip in <i>UT</i>) for 0.5 cycle 40% <i>UT</i> (60% dip in <i>UT</i>) for 5 cycles 70% <i>UT</i> (30% dip in <i>UT</i>) for 25 cycles <5% <i>UT</i> (>95% dip in <i>UT</i>) for 250ec	Complies	Mains power quality should be that of a typical commercial or hospital environment. If the user of the BRAIN MONITOR requires continued operation during power mains interruption, it is recommended that the BRAIN MONITOR to be powered from a suitable uninterruptible power supply or a battery.
Power frequency (50/60 Hz) magnetic field IEC 61000-4-8	3 A/m	Complies	Power frequency magnetic fields should be at levels characteristic of a typical location in a typical commercial or hospital environment.
NOTE: <i>UT</i> is the AC supply voltage prior to application of the test level.			

Table 3 - Electromagnetic Immunity – for EQUIPMENT and SYSTEMS that are not LIFE-SUPPORTING

Guidance and Manufacturer's Declaration – Electromagnetic Immunity			
The BRAIN MONITOR is intended for use in the electromagnetic environment specified below. The customer or the user of the BRAIN MONITOR should assure that it is used in such an environment			
Immunity test	IEC 60601 Test Level	Compliance Level	Electromagnetic Environment - Guidance
			Portable and mobile RF communications equipment should be used no closer to any part of the BRAIN MONITOR , including cables, than the recommended separation distance calculated from the equation applicable to the frequency of the transmitter.
Conducted RF IEC 61000-4-6 Radiated RF IEC 61000-4-3	3 Vrms 150 kHz to 80 MHz 3 V/m 80 MHz to 2.5 GHz	Complies	Recommended separation distance $d = 1.2 \times \sqrt{P} \quad 150\text{kHz to } 80\text{MHz}$ $d = 1.2 \times \sqrt{P} \quad 80\text{MHz to } 800\text{MHz}$ $d = 2.3 \times \sqrt{P} \quad 800\text{MHz to } 2.5\text{GHz}$ where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer and d is the recommended separation distance in meters (m). Field strengths from fixed RF transmitters, as determined by an electromagnetic site^a should be less than the compliance level in each frequency^b. Interference may occur in the vicinity of equipment marked with the following symbol: 
NOTE 1: At 80 MHz and 800 MHz, the higher frequency range applies.			
NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			
^a Field strengths from fixed transmitters, such as base stations for radio (cellular/cordless) telephones and land mobile radios, amateur radio, AM and FM radio broadcast and TV broadcast cannot be predicted theoretically with accuracy. To assess the electromagnetic environment due to fixed RF transmitters, an electromagnetic site survey should be considered. If the measured field strength in the location in which the BRAIN MONITOR is used exceeds the applicable RF compliance level above, the BRAIN MONITOR should be observed to verify normal operation. If abnormal operation is observed, additional measures may be necessary, such as re-orienting or relocating the BRAIN MONITOR .			
^b Over the frequency range 150 kHz to 80 MHz, field strengths should be less than 3 V/m.			

Table 4 - Recommended Separation Distances

Recommended Separation Distances between Portable and Mobile RF Communications Equipment and the BRAIN MONITOR.			
The BRAIN MONITOR is intended for use in an electromagnetic environment in which radiated RF disturbances are controlled. The customer or the user of the BRAIN MONITOR can help prevent electromagnetic interference by maintaining a minimum distance between portable and mobile RF communications equipment (transmitters) and the BRAIN MONITOR as recommended below, according to the maximum output power of the communications equipment.			
Rated Maximum Output Power of Transmitter (W)	Separation Distance according to Frequency of Transmitter (m)		
	150 kHz to 80 MHz	80 MHz to 800 MHz	800 MHz to 2.5 GHz
0.01	0.12	0.12	0.23
0.1	0.38	0.38	0.73
1	1.2	1.2	2.3
10	3.8	3.8	7.3
100	12	12	23
For transmitters rated at a maximum output power not listed above, the recommended separation distance d in meters (m) can be estimated using the equation applicable to the frequency of the transmitter, where P is the maximum output power rating of the transmitter in watts (W) according to the transmitter manufacturer.			
NOTE 1: At 80 MHz and 800 MHz, the separation distance for the higher frequency range applies.			
NOTE 2: These guidelines may not apply in all situations. Electromagnetic propagation is affected by absorption and reflection from structures, objects and people.			

Declaration of Compliance for FCC

Note: This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Warning: Changes or modifications not expressly approved by the manufacturer could void the user's authority to operate the equipment.

Warnings and Cautions

General Warnings

	Proper use of this device depends on the careful reading of all instructions and labels that come with or on the system. Inaccurate measurements may be caused by incorrect application or use.
	The amplifier is classified as an IP0 – ordinary degree of protection against ingress of water according to IEC 529.
	The computer used with a Brain Monitor amplifier must either be approved by XLTEK and supplied as part of an IEC 601 approved system, or it must be approved to IEC 950 or similar and kept outside of the patient environment (that is, the area within 1.5 meters of the patient laterally and within 2.5 meters of the floor in the area occupied by the patient).
	To ensure the validity of signals, do not operate the device near any sources of electromagnetic interference.
	Turn off system power before cleaning. Prevent detergent solution or cold sterilization agents from seeping into the electronics of the system. Be careful around all connectors and edges. Do not use abrasive agents.
	XLTEK systems are not AP or APG rated.
	Device accessories may include several kinds of disposable, sterile needle electrodes. These needles are labeled as STERILE , and the method of sterilization is documented on the packaging. These electrodes should not be used if the sterile packaging has been tampered with.
	The sale, distribution, or use of this device is restricted to, by, or on order of a physician.
	Printers and/or peripherals used with XLTEK devices should be marked and approved for use in their appropriate medical environment in accordance with the regulations of the country in which they are used.
	Do not attempt to connect any third-party devices to the patient breakout box except those approved by XLTEK .
	The Brain Monitor is not designed to work with, nor shall it be used with, defibrillators or electrocautery devices. The Brain Monitor could be damaged when used with these devices.
	When the Brain Monitor headbox is used in an end-use system, the system shall be evaluated using IEC 60601-1-1.

	Equipment used in the patient environment shall be IEC 601-1 approved. Equipment used outside the patient environment shall be at least approved to IEC 950 standard.
---	---

Electrical Warnings and Cautions

	XLTEK systems are intended for connection to a properly grounded electrical outlet only.
	ELECTRICAL SHOCK HAZARD: Do NOT turn on the system power until all cables have been connected, verified and visually inspected for any damage. Failure to inspect the cables may result in electrocution.
	ELECTRICAL SHOCK HAZARD: Do NOT connect electrode inputs to earth ground. The patient breakout box contains warning symbols reminding you that the connections are intended for isolated patient connections only. Connecting to an earth ground might result in electrocution.
	ELECTRICAL SHOCK HAZARD: Do NOT service the system. Refer servicing to qualified personnel only.

Patient Environment Warnings and Cautions



NOTE: The patient environment is defined as the area within 1.5 meters of the patient laterally and within 2.5 meters of the floor in the area occupied by the patient.

	Connect all patient electrodes to fully electrically isolated devices only. Connecting patient electrodes to any other device or external outlet may result in personal injury.
	The patient breakout box accepts only touch-proof style electrode inputs. Do NOT attempt to use any other style of patient electrode input.
	The patient event switch attached to the Brain Monitor base unit is NOT intended for critical patient-safety-related incidents.
	Patient connections are NOT intended for direct cardiac contact.
	Route patient cabling carefully to reduce the possibility of patient entanglement or strangulation.

Masimo Pulse Oximeter Warnings

	The Masimo SET pulse oximeter is to be operated by qualified personnel only. This manual, accessory directions for use, all precautionary information, and specifications should be read before use.
	A pulse oximeter is intended for diagnostic purposes and should NOT be used as an apnea monitor.
	A pulse oximeter should be considered an early warning device . As a trend towards patient deoxygenation is indicated, blood samples should be analyzed by a laboratory co-oximeter to completely understand the patient's condition.
	Significant levels of dysfunctional hemoglobins (HbCO or MetHb) may cause inaccurate measurements.
	Intravascular dyes such as indocyanine green or methylene blue may cause inaccurate measurements.
	Excessive illumination may cause inaccurate measurements or loss of pulse signal.
	Excessive patient movement may cause inaccurate measurements.
	Venous pulsations may cause inaccurate measurements.
	The board may be used during defibrillation, but the readings may be inaccurate for a short time.
	Carboxyhemoglobin may erroneously increase readings. The level of increase is approximately equal to the amount of carboxyhemoglobin present. Dyes, or any substance containing dyes that change usual arterial pigmentation, may cause erroneous readings.
	Loss of pulse signal can occur when the patient has hypotension, severe vasoconstriction, severe anemia, or hypothermia.
	Loss of pulse signal can occur when there is arterial occlusion proximal to the sensor.
	Loss of pulse signal can occur when the patient is in cardiac arrest or is in shock.
	Tissue damage can occur due to incorrect placement of sensor.
	Do not immerse the sensor in water, solvents, or cleaning solutions. Do not sterilize the sensor by irradiation, steam, or ethylene oxide.

	Do not immerse patient cables in water, solvents, or cleaning solutions (the patient cable connectors are not waterproof). Do not sterilize patient cables by irradiation, steam, or ethylene oxide.
---	--

Masimo Pulse Oximeter Sensor Warnings

	Before use, carefully read the LNOP sensor directions for use.
	Use only Masimo oximetry sensors for SpO ₂ measurements. Other oxygen transducers (sensors) may cause improper pulse oximeter performance.
	Tissue damage can be caused by incorrect application or use of an LNOP sensor; for example, by wrapping the sensor too tightly. Inspect the sensor site as directed in the sensor directions for use to ensure skin integrity and correct positioning and adhesion of the sensor.
	Do not use Masimo oximetry sensors during MRI scanning as this could potentially cause burns.
	Placement of a sensor on an extremity with a blood pressure cuff, arterial catheter, or intravascular line may cause inaccurate measurements.
	Loss of pulse signal can occur when the sensor is too tight.
	Do not use damaged LNOP sensors. Do not use an LNOP sensor with exposed optical components. Do not immerse the sensor in water, solvents, or cleaning solutions (the sensors and connectors are not waterproof). Do not sterilize by irradiation, steam, or ethylene oxide. See the cleaning instructions in the directions for use for reusable Masimo LNOP sensors.
	Do not use damaged patient cables.

Description of Symbols

Symbol	Description
	ATTENTION: Consult Accompanying Documents
	Consult Accompanying Documents
	Protective Earth (Ground)
	Type BF Equipment
	Dangerous Voltage
	Alternating Current
	Direct Current
	Power On
	Power Off
	EU only: Do Not Dispose as Unsorted Municipal Waste
	Class II Equipment (non-grounded enclosure)
	ESD Sensitive or Static Sensitive

Specifications: Brain Monitor Amplifier

Specification	Value(s)
Data Transmission	
Transmission Type	Physical Connection
Analog Specifications Channels	
Referential Channels	32
Differential Channels	6
Differential Input Impedance	> 100 MOhms
Common Mode Input Impedance	> 50 MOhms
Common Mode Rejection Ratio	- 112 dB @ 60 Hz
Input Noise (peak to peak)	2.4 μ V @ 0.1Hz~70Hz bandwidth
Input Noise (RMS)	0.39 μ V @ 0.1Hz~70Hz bandwidth
Maximum Differential AC Input Before Clipping (Referential)	20 mV pk-to-pk
Maximum Differential AC Input Before Clipping (Differential)	40 mV pk-to-pk
Maximum Operational DC Input Voltage Electrode Offset	+/-200 mV
Input Bias Current	< 20 pA
Channel Crosstalk	< -67 dB
Analog Specifications – DC Channels	
DC Channels (patient side)	4 non-isolated
DC Channels (computer side)	6 isolated
DC Input Range	+/-5 V
Photic Stim Interface	
Photic Stim Output	1 (TTL level, active high)
Photic Stim Input	1 (TTL level, active high)

Brain Monitor Amplifier Specifications, cont.

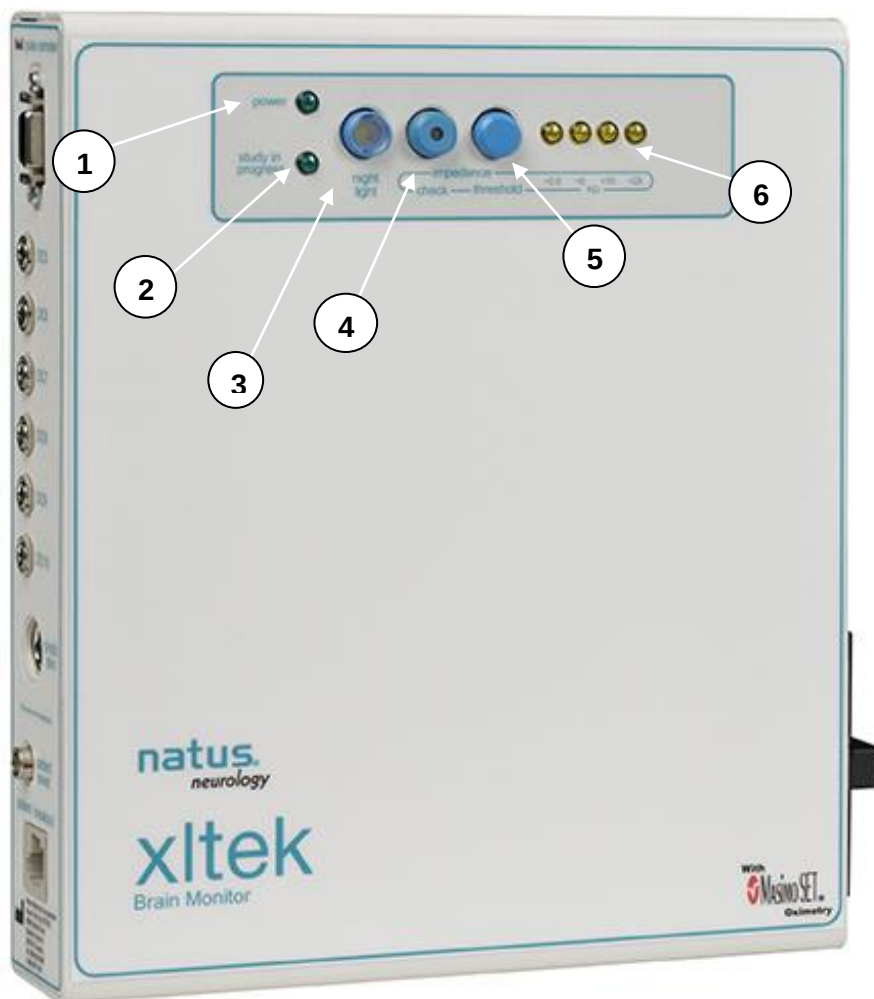
User Interface	
Push Buttons	3 (Nightlight, Impedance, Impedance Threshold)
Event Switch Connection	1
LED Indication	5 on main unit; 43 dual color on breakout box.
Night Light	White LEDs in breakout box
Digital Specifications	
Sampling Frequency	200 Hz, 256 Hz, 512 Hz
Sampling Resolution – Referential Channels	16 bits
Sampling Quantization – Referential Channels	0.3 μ V
Sampling Resolution - Differential Channels	16 bits
Sampling Quantization – Differential Channels	0.6 μ V
Sampling Resolution - DC Channels	16 bits
Sampling Quantization - DC Channels	0.2 mV
Modes of Operation	
Power	6 W @120 V
Impedance Check	< 2.5, < 5, < 10, < 25 kOhms
Channel Test Signal	10 Hz, 650 μ V pk-to-pk
Host Interface	
Ethernet 100BaseT (Option 1)	TCP/IP, UDP, DHCP
USB 2.0 (Option 2)	USB 2.0 High Speed.
Headbox Mechanical	
Interface Cable (Option 1)	RJ45 cable
Interface Cable (Option 2)	USB 2.0 cable, maximum 2 meters
Headbox to Breakout Box Cable	Standard Length: 52 inches (123 cm) Maximum Length: 30 feet (~ 12 m)
Approx. Size (inches)	Main: 7.5 x 8 x 1.7 Breakout: 6.1 x 3.8 x 1.1

Brain Monitor Amplifier Specifications, cont.

Environmental Conditions for Use	
Operating Environmental Limits	Temperature: 10° to 40°C
	Relative Humidity: 30%–75%
	Atmospheric Pressure: 700 hPa–1060 hPa
Transport and Storage Temperature Range	– 40°C to 70°C
Transport and Storage Humidity Range	10–100%, including condensation
Transport and Storage Atmospheric Pressure Range	500 hPa to 1060 hPa
General	
Electrical Ratings	115/230 V AC, 50/60 Hz, 20 W
Fuse Type and Rating	T 0.5A / 250 V

Product Images

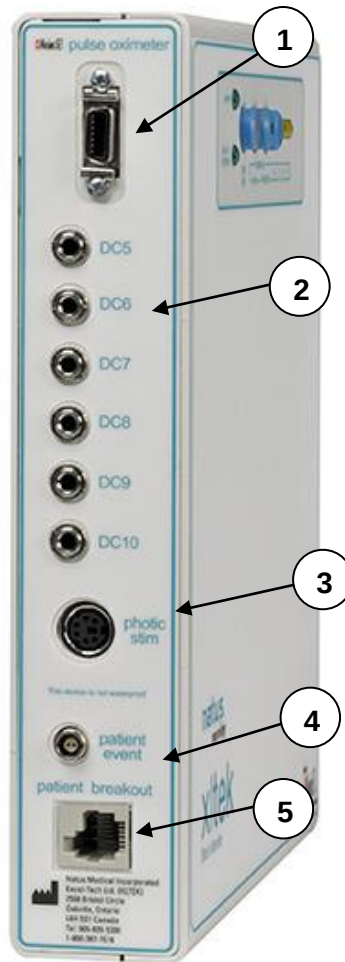
Brain Monitor Base Unit



Front View of Brain Monitor Base Unit

Brain Monitor Base Unit – Front	
1	Power Indicator
2	Status Indicator
3	Activate Night Light Button
4	Start Impedance Test Button
5	Change Impedance Threshold Button
6	Impedance Threshold Indicators

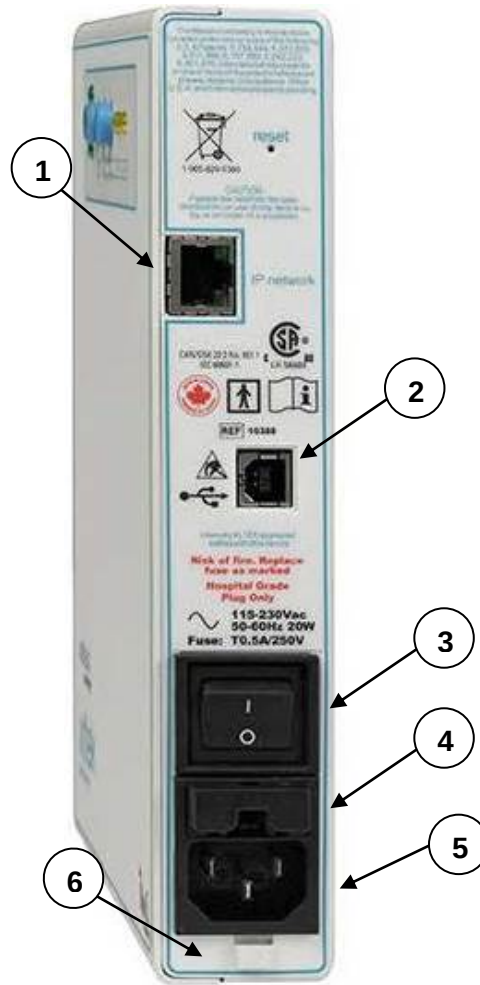
Brain Monitor Base Unit (Left Side)



Left Side of Brain Monitor Base Unit

Brain Monitor Base Unit – Left Side	
1	 Pulse Oximeter Connection
2	DC Inputs
3	Photic Stim Connection
4	Patient Event Connection
5	Breakout Box Connection

Brain Monitor Base Unit (Right Side)



Right Side of Brain Monitor Base Unit

Brain Monitor Base Unit – Right Side	
1	IP Network Connection
2	USB Connection
3	On/Off Switch
4	Fuse Compartment
5	Power Source Connection
6	Power Cord Release Button

Brain Monitor Breakout Box



Brain Monitor Breakout Box

Brain Monitor Breakout Box	
1	Standard 10-20 Channel Inputs
2	Base Unit Connection
3	Additional Referential Inputs
4	Differential Inputs
5	DC Inputs

Unpacking

When you unpack your Brain Monitor, make sure the following items are included:

- Brain Monitor base unit
- Brain Monitor breakout box
- Brain Monitor breakout cable
- Ethernet cable
- Power cord
- Patient event switch

Optional Accessories:

- Sleep accessory kit
- EEG accessory kit



NOTE: The Brain Monitor amplifier should be used only with cables, transducers, electrodes, sensor, and switches that are supplied or approved by **XLTEK**.

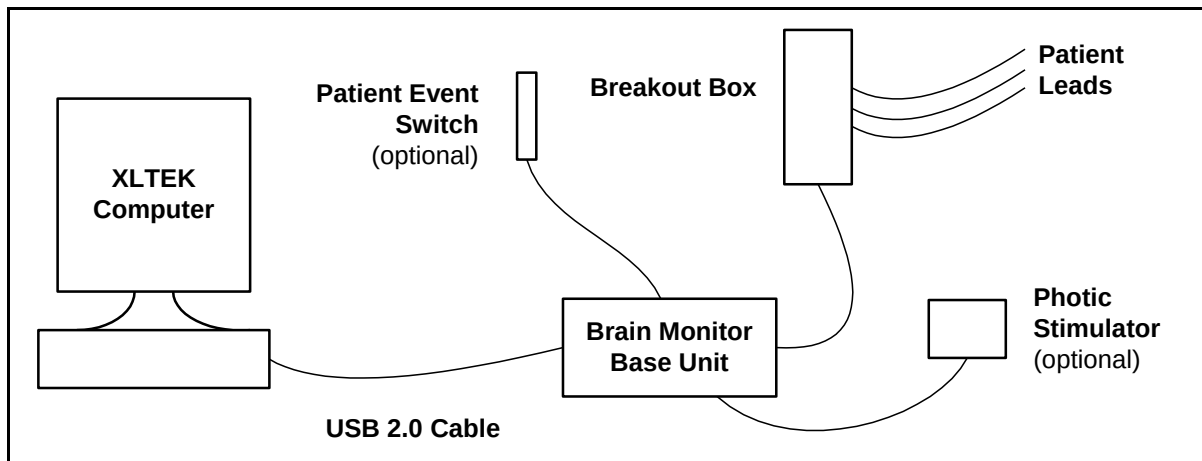
Setting Up

The Brain Monitor amplifier is designed to work with an **XLTEK** computer system running XLTEK Database (XLDB) and NeuroWorks or SleepWorks software.

USB-Style Connection

To connect the Brain Monitor to the **XLTEK** computer:

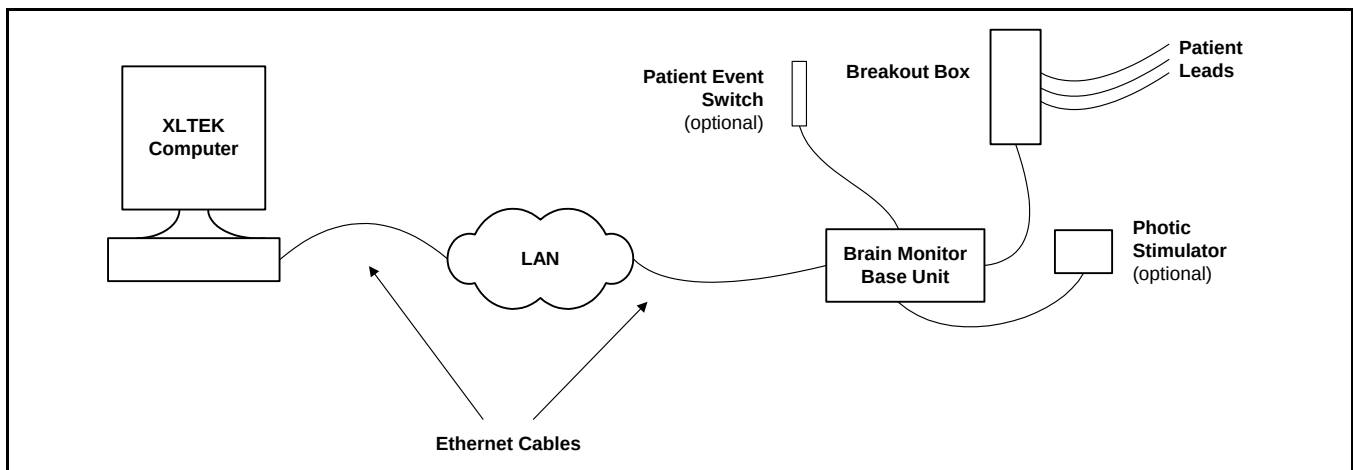
1. Use the supplied USB 2.0 cable to connect the Brain Monitor base unit to the computer.
2. Insert the Patient Event Switch into the Patient Event connection on the base unit.
3. If a Photic Stimulator is needed, insert the Photic Stimulator into the Photic Stimulator connection on the base unit.
4. When you are ready to run a study, connect the patient leads and transducers to the Brain Monitor.



USB-Style Hardware Connections for a Brain Monitor Amplifier

Ethernet-Style Connection

1. Connect the computer to the LAN (Local Area Network).
2. Connect the Brain Monitor base unit to the LAN.
3. Insert the Patient Event Switch into the Patient Event connection on the base unit.
4. If a Photic Stimulator is needed, insert the Photic Stimulator into the Photic Stimulator connection on the base unit.
5. When you are ready to run a study, connect the patient leads and transducers to the Brain Monitor.



Ethernet-Style Hardware Connections for a Brain Monitor Amplifier



NOTE: **XLTEK** designed the Brain Monitor amplifier for EEG and Sleep (polysomnography) studies. For more information on running studies, refer to the online **Help**.

Testing the Brain Monitor Amplifier

The Brain Monitor amplifier is fully assembled, tested, and calibrated before being shipped to you. The following sections describe some tests you can perform to verify the performance of the Brain Monitor amplifier.

Calibration and Verification

There is no need to calibrate the software or the Brain Monitor amplifier. All calibration is done at the factory before the system is shipped. To verify that the Brain Monitor amplifier is correctly calibrated, perform the following procedure:

1. Connect the Brain Monitor base unit to an **XLTEK** computer and turn on the system.
2. Start XLTEK Database (XLDB).
3. To start a new study, click **New EEG** or **Sleep**.

4. Choose **Edit > Settings > Acquisition**.
5. On the Acquisition tab, set the **Reference Electrode** to **Common**.
6. Design four bipolar montages that take the difference of adjacent channels; for example, C3-CZ, C4-T4, T5-P3, etc.
7. Apply a sine wave of 50 microvolts, peak-to-peak amplitude, 10 Hz to all channels of the group using a signal generator. Ensure there is a 50 Ohm load on the generator output if the generator is designed to deliver the specified level into this load.
8. Set the LFF filter to 0.1, the HFF filter to OFF, and the Notch filter to OFF.
9. Verify that no sine wave is greater than 50 microvolts peak-to-peak. 50 microvolts represents gain match to 1%.



NOTE: For more information on setting up a montage, consult the online Help.

Channel Test

While in **Acquisition** mode, a channel test may be performed to check whether a signal is being properly processed from the amplifier to the display. A channel test applies a **test signal** to all channels. This allows you to examine the waveforms on the screen to see if all channels are working.



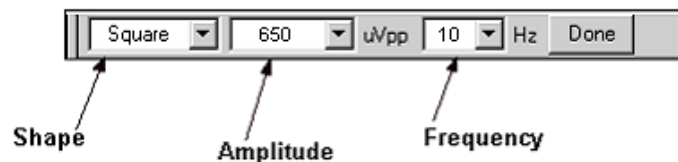
NOTE: A channel test does not validate the connection from the patient electrode to the amplifier input.

To Run a Channel Test

1. Choose **Controls > Channel Test Signal**. The **Channel Test** toolbar appears above the waveform window.
2. Use the Channel Test toolbar to select the desired wave **shape**, **amplitude**, and **frequency**.
3. To stop the channel test and save the current settings, click **Done**.

Channel Test Signal Control

The Channel Test Signal control turns on the channel test signal according to the last settings saved and displays a **Test Signal** toolbar. The Test Signal toolbar is located above the trace display. The toolbar has controls for shape, amplitude and frequency.



Channel Test Signal Toolbar

Impedance Check

An impedance check is performed to ensure that the electrode contact with the patient is satisfactory. You can perform an impedance check at any time during a study.

To Run an Impedance Check from the Software

When an impedance check is initiated, the software scans all channels (in auto scan mode).

To...	Do this...
Start the impedance check	Choose Controls > Impedance Check
Lock onto a channel	Click Lock Channel . Then, make adjustments to the electrode connection until satisfactory levels are achieved
Proceed to a full impedance check	Click Release Lock
End the impedance check	Click End
Save the impedance check as part of the study	Click End and Start Recording

An impedance check displays bar graphs that show the impedance of each electrode connection. A **green** bar indicates that the reading is below the set threshold. A **red** bar indicates the reading is above the threshold. To set the impedance threshold, click one of the **Threshold** buttons in the **Threshold Group** box on the right side of the **Check Impedance** box.

Pulse Oximeter

The following topics list the specifications cleared by the FDA under K990966 for Masimo SET board that is built into the Brain Monitor. The board connects to sensors and provides oxygen saturation, pulse rate, pulse waveform, and other output information via the SleepWorks software. The Brain Monitor provides power and isolates the board from the main power and ground.

Masimo Patents

This device is covered under one or more of the following U.S. Patents: 5,482,036; 5,490,505; 5,632,272; 5,685,299; 5,758,644; 5,769,785; 6,002,952; 6,036,642; 6,067,462; 6,206,830; 6,157,850.

No Implied License

Possession or purchase of this device does not convey any express or implied license to use the device with replacement parts which would, alone, or in any combination with this device, fall within the scope of one or more of the patents relating to this device.

Indications for Use

The pulse oximeter and accessories are indicated for the continuous noninvasive monitoring of functional oxygen saturation of arterial hemoglobin (SpO₂) and pulse rate (measured by an SpO₂ sensor). The Masimo SET pulse oximeter and accessories are indicated for use with adult, pediatric, and neonatal patients during both no motion and motion conditions, and for patients who are well or poorly perfused in hospitals, hospital-type facilities, mobile, and home environments.

How the Pulse Oximeter Works

The pulse oximeter is based on three principles:

1. Oxyhemoglobin and deoxyhemoglobin differ in their absorption of red and infrared light (spectrophotometry).
2. The volume of arterial blood in tissue (and hence, light absorption by that blood) changes during the pulse (plethysmography).
3. The arterio-venous shunting is highly variable, and that fluctuating absorbance by venous blood is a major component of noise during pulse. Because oxyhemoglobin and deoxyhemoglobin differ in light absorption, the amount of red and infrared light absorbed by the blood is related to hemoglobin oxygen saturation.

The pulse oximeter decomposes the red and infrared pulsatile absorbance signal into an arterial signal plus a noise component and calculates the ratio of the arterial signals without noise. When the ratio of the two arterial pulse-added absorbance signals is calculated, its value is used to find the (SpO₂) saturation employing an empirically derived equation in the pulse oximeter's software. The values in the look-up table are based upon human blood studies using a laboratory co-oximeter on healthy adult volunteers in induced hypoxia states during motion and non-motion conditions.

The pulse oximeter, as well as traditional pulse oximetry, determines SpO₂ by passing red and infrared light into a capillary bed and measuring changes in light absorption during the pulsatile cycle. Red and infrared light-emitting diodes (LEDs) in oximetry sensors serve as the light sources; a photodiode serves as the photodetector.

Traditional pulse oximetry assumes that all pulsations in the light absorbance signal are caused by oscillations in the arterial blood volume. This assumes that the blood flow in the region of the sensor passes entirely through the capillary bed rather than through any arterio-venous shunts. The traditional pulse oximeter calculates the ratio of pulsatile absorbance (AC) to the mean absorbance (DC) at each of two wavelengths, 660 nm and 940 nm:

$$S(660) = AC(660)/DC(660)$$

$$S(940) = AC(940)/DC(940)$$

The oximeter then calculates the ratio of these two arterial pulse-added absorbance signals:

$$R = S(660)/S(940)$$

This value of R is used to find the saturation SpO_2 in a look-up table built into the oximeter's software. As stated previously, the values in the look-up table are based upon human blood studies using a laboratory co-oximeter on healthy adult volunteers in induced hypoxia studies.

The pulse oximeter assumes that arterio-venous shunting is highly variable and that fluctuating absorbance by venous blood is the major component of noise during the pulse. The pulse oximeter decomposes S(660) and S(940) into an arterial signal plus a noise component and calculates the ratio of the arterial signals without the noise:

$$S(660) = S1 + N1$$

$$S(940) = S2 + N2$$

$$R = S1/S2$$

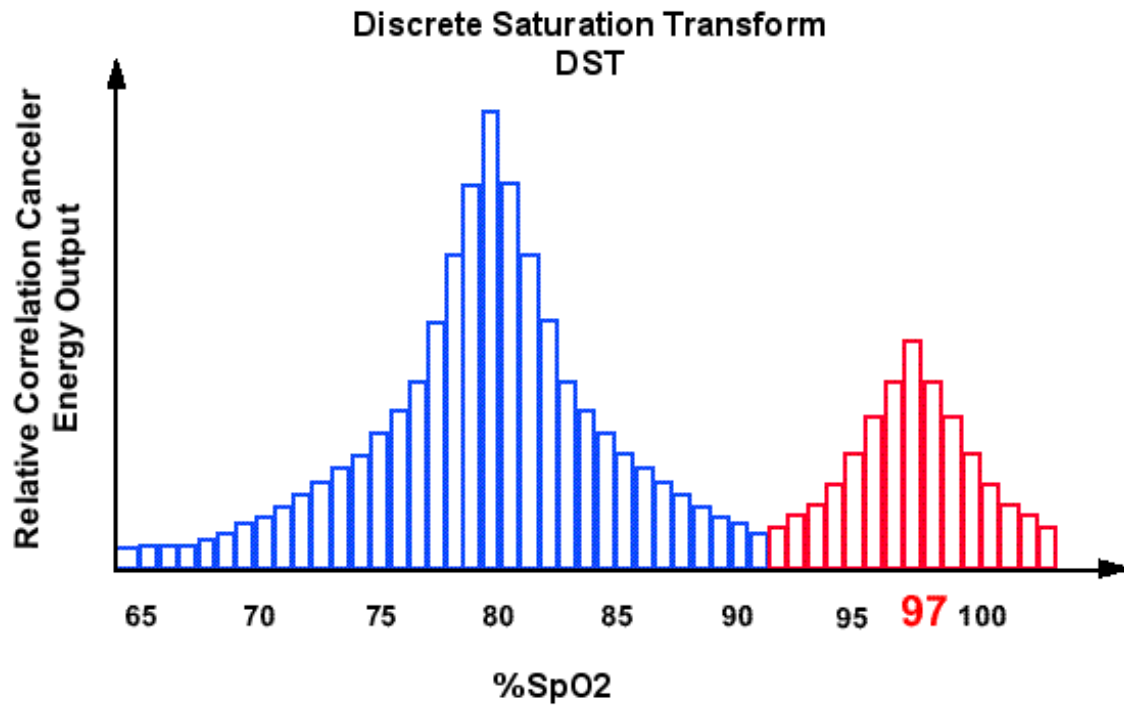
Again, R is the ratio of two arterial pulse-added absorbance signals, and its value is used to find the saturation SpO_2 using an empirically derived equation in the oximeter's software.

The above equations are combined and a noise reference (N') is determined:

$$N' = S(660) - S(940) \times R$$

If there is no noise $N' = 0$: then $S(660) = S(940) \times R$ which is the same relationship for the traditional pulse oximeter.

The equation for the noise reference is based on the value of R, the value being sought to determine the SpO_2 . The pulse oximeter software sweeps through possible values of R that correspond to SpO_2 values between 1% and 100% and generates an N' value for each of these R values. The S(660) and S(940) signals are processed with each possible N' noise reference through an adaptive correlation canceller (ACC) which yields an output power for each possible value of R (i.e. each possible SpO_2 from 1% to 100%). The result is a Discrete Saturation Transform (DST) plot of relative output power versus possible SpO_2 value as shown in the following figure where R corresponds to $SpO_2 = 97\%$.



The DST plot has two peaks: the peak corresponding to the higher saturation is selected as the SpO₂ value. This entire sequence is repeated once every two seconds on the most recent four seconds of raw data. The pulse oximeter SpO₂, therefore, corresponds to a running average of arterial hemoglobin saturation that is updated every two seconds.

Pulse Oximeter Accessories

PC Series Cable: Connects the pulse oximeter module to the oximetry sensors. Transfers LED drive power to the oximetry sensors from the pulse oximeter. The pulse oximeter receives the detector signal from the oximetry sensor through the PC series cable.

LNOP Series Oximetry Sensors: Measure the light absorption of blood from two light emitting diodes (LED's). Oxygen saturated blood absorbs light differently compared to unsaturated blood. The amount of light absorbed by the blood is used to calculate the ratio of oxygenated hemoglobin to total hemoglobin in arterial blood.

Available Oximetry Sensors

Single Use Oximetry Sensor	For adults and pediatrics greater than 30 kg
	For pediatrics and small adults greater than 10 kg and less than 50 kg
	For neonates less than 10,000 grams
	For neonates less than 1,000 grams
Reusable Oximetry Sensor	For adults and pediatrics greater than 30 kg

Measurements

If the accuracy of any measurement does not seem reasonable, check the patient's vital signs by alternative means. Then, check the pulse oximeter for proper functioning.

Causes of Inaccurate Measurements

- Incorrect sensor application or use.
- Significant levels of dysfunctional hemoglobins (e.g. carboxyhemoglobin or methemoglobin).
- Intravascular dyes such as indocyanine green or methylene blue.
- Exposure to excessive illumination, such as surgical lamps (especially ones with a xenon light source), bilirubin lamps, fluorescent lights, infrared heating lamps, or direct sunlight. Exposure to excessive illumination can be corrected by covering the sensor with a dark or opaque material.

Causes of Loss of Pulse Signal

- Sensor is too tight.
- Excessive illumination from light sources such as a surgical lamp, a bilirubin lamp, or sunlight.
- Blood pressure cuff is inflated on the same extremity as the one with a SpO₂ sensor attached.
- Patient has hypotension, severe vasoconstriction, severe anemia, or hypothermia.
- Arterial occlusion is proximal to the sensor.
- Patient is in cardiac arrest or is in shock.

Maintenance and Cleaning

To keep the Brain Monitor amplifier in good working condition, follow a regular schedule of preventive maintenance. Regular preventive maintenance does not involve access to the interior of the Brain Monitor amplifier and components. For servicing problems that require corrective maintenance and/or internal component service, call **Natus** Technical Support at **1-800-303-0306**, or contact your local **XLTEK** representative.

Periodically check cable connections and electrodes for damage and wear. Inspect cables for bent pins. Replace frayed or worn cables. Also, regularly inspect and clean all system components, including:

- Connectors and jack ports
- Base unit, breakout box and USB 2.0 cable
- Electrodes and accessories

Like any piece of equipment, the Brain Monitor is subject to increased daily wear and tear. Taking basic care of the system, and avoiding extreme physical abuse, helps prolong the lifespan of the Brain Monitor.

Do's and Don'ts

	Disconnect all cables from the Brain Monitor before wiping. Use a lint-free cloth. Do not use cleaners on any system component.
	Be careful not to allow any fluid to seep into the internal electronic components of the system.
	Do NOT leave the amplifier attached to the computer when transporting the unit.
	Do NOT autoclave, pressure sterilize, or gas sterilize this amplifier.
	Do NOT soak or immerse the amplifier in any liquid.
	A cleaning solution of 70% isopropyl alcohol is recommended.
	Use cleaning solution sparingly. Excessive solution can flow into the amplifier and cause damage to internal components.
	Do NOT use petroleum-based or acetone solutions, or other harsh solvents, to clean the amplifier.

Troubleshooting

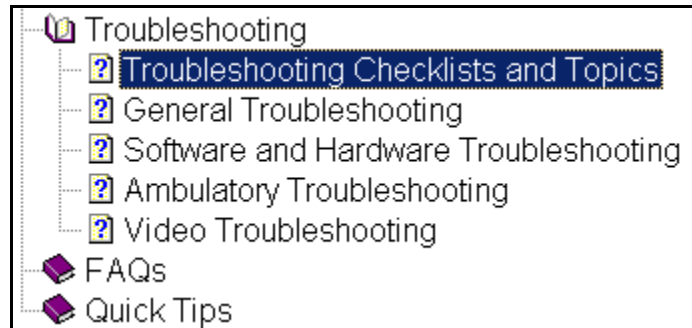
If you are experiencing problems, try the solutions listed below.

Troubleshooting Checklist

	Ask the patient to relax.
	Inspect your cables.
	Make sure all cables are connected properly between the Brain Monitor and the computer. If using a LAN, make sure the hardware and software are configured correctly.
	Make sure the patient electrodes are connected to the correct channels in the breakout box.
	Make sure the patient electrodes fit properly into the breakout box (not loosely).
	Make sure there are no apparent breaks in the patient electrode cables.
	Are any of the electrodes touching? If so, they are causing a short circuit and will develop an artifact.
	Check the impedance.
	Unplug any other devices on the same circuit such as printers, mechanical beds, vacuum cleaners, or other potential sources of interference.
	Check the gain and timebase settings to ensure that they are appropriate for the current test. You may also want to check the LFF, HFF, and Notch Filter settings. Choose Edit > Settings > Montage . Right-click a cell in the appropriate column and select a value.
	Turn the base unit off and on. This will reset the system.
	Shut down for at least 10 seconds. Then set up the test again from the beginning. Shutting down the computer and restarting sometimes solves the problem.

Opening the Troubleshooting Section of the Online Help

1. In NeuroWorks, choose **Help > XLTEK NeuroWorks EEG Help**.
2. Click the **Contents** tab.
3. Double-click **NeuroWorks**.
4. Double-click the **Troubleshooting** book. The Troubleshooting book expands to show the contents.
5. Click a topic to display the content in the topic pane on the right.



Troubleshooting Section of the NeuroWorks Online Help

Getting Help

DBAXL TEK is committed to providing you with support so you can operate the Brain Monitor amplifier with ease and confidence. If you need help, follow these steps to find a solution:

Step 1: Document the Incident

Carefully document the incident. If possible, note error messages, dialog box names and what you did before the problem occurred.

Step 2: Search the Online Documentation

Choose:

- **Help > Headboxes**

Step 3: Restart the Computer

Often restarting the computer will solve a problem.

1. Close all applications.
2. Click the **Start** button on the Windows taskbar.
3. Choose **Shut Down** from the Start menu.
4. Select **Restart the computer** and click **Yes**.

Step 4: Shut Down the Computer

Sometimes you need to shutdown the computer completely in order to solve a problem.

1. Click the **Start** button on the Windows taskbar.
2. Choose **Shut Down** from the Start menu.
3. Select **Shut Down** and click **Yes**.
4. Turn off the power to the unit. Wait 10 seconds. Turn the power back on.

Step 5: Contact Technical Support

First, write down the serial number of your computer (located on the back) and the serial number of your Brain Monitor amplifier. Then contact your local XLTEK distributor or Technical Support at **1-800-303-0306** or Oakville_Technical_Service@natus.com.

We welcome your feedback and suggestions regarding the Brain Monitor amplifier and any aspect of our SleepWorks or NeuroWorks systems and software, online Help, line of accessories, and support services.

EEG and Sleep Accessories

EEG and Sleep accessories which can be used with the Brain Monitor amplifier are available for you to browse in the Natus Neurology Accessories Catalog online at www.natus.com or call Natus Sales and Support 1-800-303-0306.

Intentionally Left Blank

Intentionally Left Blank



A Total Service Solution

Standing behind every XLTEK product is Natus Medical Incorporated, an internationally respected innovator of medical products and services.

Our Neurology systems are backed up by an in-house support team staffed with technical and clinical experts, 24/7 support, remote support via WebEx or VPN, the largest clinical and technical field support network in Neuro/Sleep and customized service contracts that include preventative maintenance visits and computer upgrades.

Natus Medical Incorporated
DBA Excel-Tech Ltd. (XLTEK)
2568 Bristol Circle
Oakville, Ontario
L6H 5S1 Canada
T: 905.829.5300
Toll Free (US & Canada): 800.303.0606
www.natus.com

P/N 105124 Rev I